

How to Develop Infrastructure Platform Engineering Teams

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Initiatives: Functional Design

Infrastructure platform engineering is the future of I&O, but it is difficult to implement and requires an exceptional team with dedicated resources. To build successful IPE teams, I&O leaders must focus on a new organizational structure, and new roles and skills.

More on This Topic

This is part of an in-depth collection of research. See the collection:

- [Research Roundup for DevOps](#)

Overview

Key Findings

- Traditional infrastructure and operations (I&O) team structure, skills and roles do not enable infrastructure platforms delivery, forcing I&O leaders to rethink their organizational and staffing strategy to deliver agile infrastructure platforms.
- Successful infrastructure platform engineering (IPE) implementations require agile software engineering capabilities and practices that many I&O teams don't have.
- Competition is fierce for skilled staff, and I&O leaders struggle to attract and develop talent necessary to adopt a successful infrastructure platform engineering practice.

Recommendations

- Prioritize infrastructure platform engineering by investing in dedicated resources that operate as an agile product team with the required skills to deliver customer-centric platform capabilities.
- Create the necessary roles and invest to acquire platform engineering skills, including hiring software engineers with a passion for IPE and partnering with external providers.
- Develop a comprehensive strategy including hiring, training, closing skills gaps and partnering with service providers.

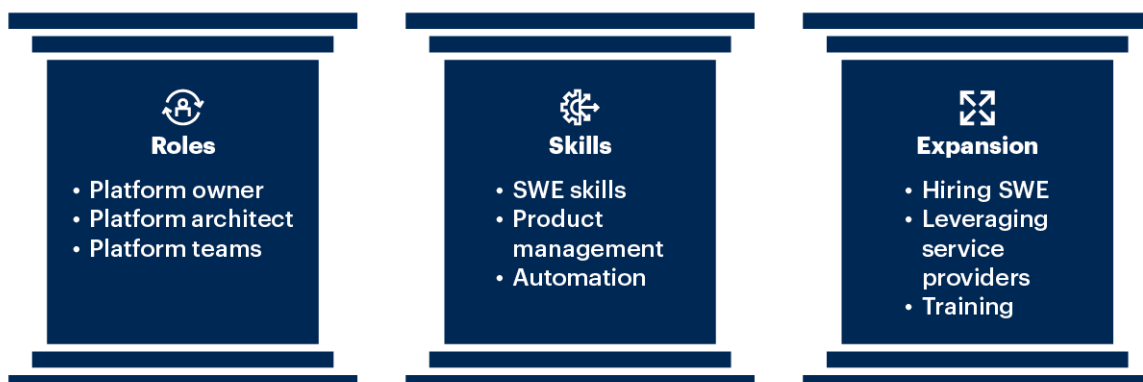
Introduction

A commitment to operational excellence alone is not enough for I&O leaders delivering infrastructure services. They must improve velocity and deliver infrastructure platform capabilities that meet the agility needs of their organization. Investing in building a successful infrastructure platform engineering (IPE) team to meet these demands has become a critical priority. Applying IPE to deliver and operate infrastructure platforms is not easy; it requires new roles, new team structure and new skills that many I&O teams don't have.

This research highlights the key elements needed to build a successful IPE team, including roles, skills and expanding resources (see Figure 1).

Figure 1: Building a Successful IPE team

Building a Successful IPE team



Source: Gartner
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Analysis

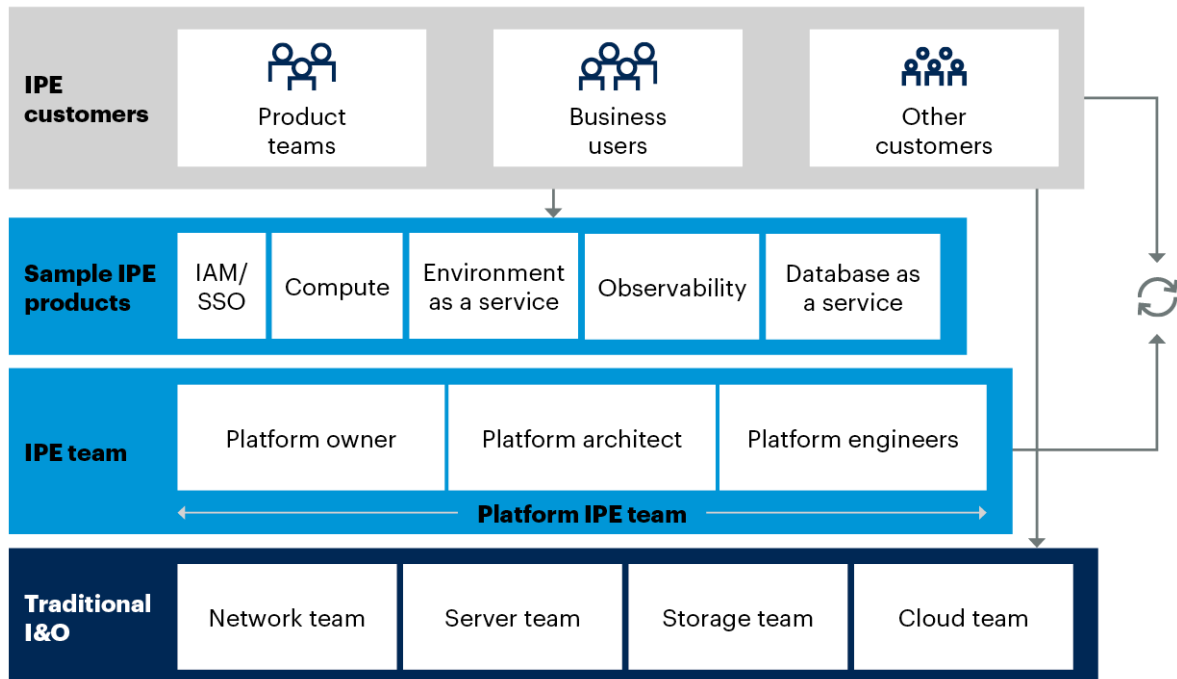
Invest in Dedicated Resources and Roles

Traditional I&O is not structured to deliver end-to-end platform-centric capabilities. It is organized by technology silos working independently, involving multiple engineers from multiple teams, requiring many handoffs and causing user frustration.

The I&O leader must invest in dedicated resources that will be accountable for provisioning and operating platform capabilities. Figure 2 shows IPE teams, sample products and customers. The IPE team has all the required roles to deliver the platform capabilities. The IPE resources work as a product team and are accountable as a group to deliver.

An IPE team is an agile product development team and is accountable for delivery and operations of infrastructure platforms and tools. The IPE team will apply product management and software engineering practices to deliver and evolve customer-driven capabilities.

Figure 2: Infrastructure Platform Team Organizational Positioning

Infrastructure Platform Team Organizational Positioning

Source: Gartner
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Gartner

Delivering and sustaining modern infrastructure platform capabilities as products by leveraging software engineering practices is not easy for I&O teams. These skills are normally associated with application development teams.

Platform capabilities need to be demand-driven. They start with the thinnest viable platform and continue to evolve to meet customer needs. The IPE team needs to minimize the level of effort to consume the delivered capabilities, regardless of their customers' level of expertise. The effectiveness and success of the IPE team require it to operate like a product team (see [Key Decisions When Forming Software Engineering Teams](#)). This includes all engineering and operational aspects, marketing the platform(s), including soliciting and analyzing data on usage and sentiment, and collaboration from the broader community.

The infrastructure platform engineering team is composed of three major roles:

- Platform owner
- Platform architect

■ Platform engineer

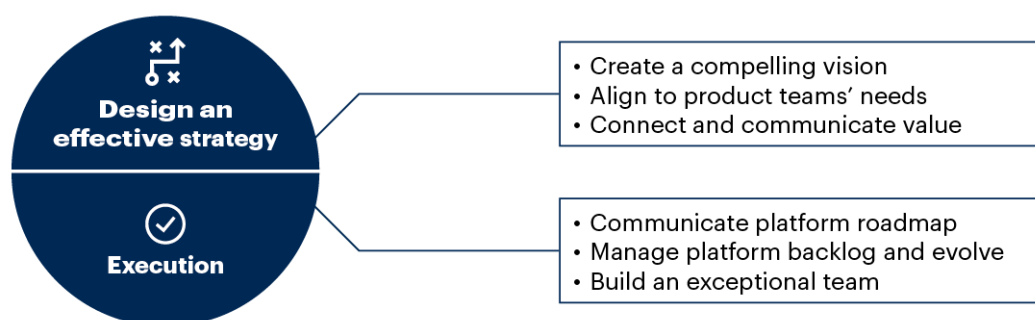
Platform Owner

The platform owner fulfills the role of product owner, and is responsible for the platform products and for meeting its customers' needs. For this role, I&O leaders must select an individual with a software engineering product management background. They need to possess the right mix of skills, including "people" skills (such as communication and collaboration) and business acumen. The platform owner will be responsible for ensuring the team operates as an agile software delivery team and must provide the necessary training and coaching.

There are two broad sets of responsibilities for infrastructure platform owners: strategy and execution. See Figure 3 for sample responsibilities.

Figure 3: Infrastructure Platform Owner Sample Skills

Infrastructure Platform Owner Sample Skills



Source: Gartner
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Platform owners are key to success, as they set the strategy and vision, and manage platform quality to meet customer demands. They need to be highly engaged with customers such as software developers or business technologists, and prioritize new platform capabilities and enhancements based on customers' needs.

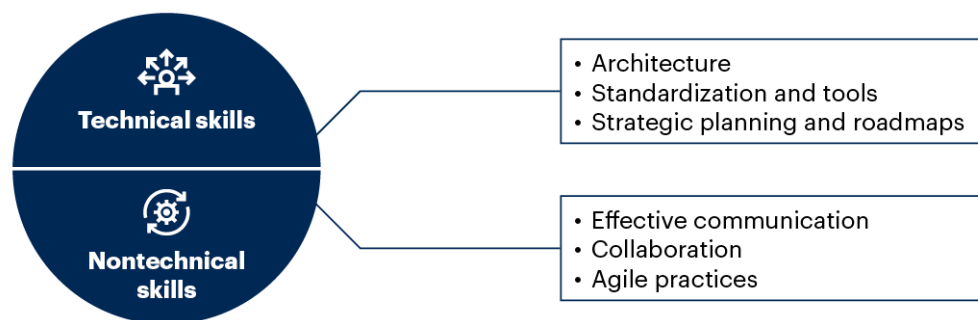
After creating the strategy, the infrastructure platform owner must execute on it. This includes communicating the infrastructure platform roadmap to the customers that can/should consume the services. The platform owner needs to evolve platforms based on adoption, feedback and new requirements, and must demonstrate success through metrics such as customer satisfaction, feature usage, ease of use and reliability.

Platform Architect

The infrastructure platform architect must connect the IPE team to platform customers. This role helps set standards and best practices, and interacts with customers to drive customer requirements in the infrastructure roadmap and planning. Initially, the role of platform architect may be fulfilled by the platform owner or federated among the IPE team. Over time, there may be a need for a dedicated role to support the services delivered. This role requires a mix of technical and nontechnical responsibilities and skills (see Figure 4).

Figure 4: Infrastructure Platform Architect Sample Skills

Infrastructure Platform Architect Sample Skills



Source: Gartner
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The platform architect is responsible for developing and governing the architectures associated with the infrastructure platforms. This role also develops standards for tooling. Architects lead the strategic infrastructure planning activities and the development of roadmaps.

This role also must collaborate with key stakeholders from the development community, business groups and other infrastructure colleagues. Working with agile practices is a key aspect of modern infrastructure platforms. It requires iterative delivery with continuous feedback for continuous improvements.

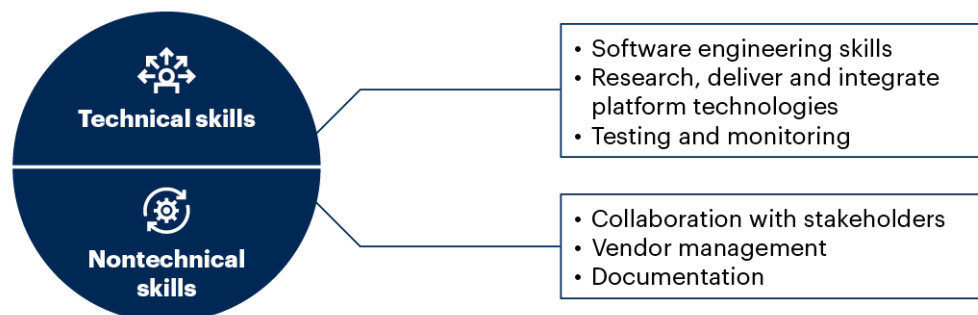
Platform Engineer

Infrastructure platform engineers are responsible for the delivery and operations of the platform products. The size of the IPE team will depend on the initial planned capabilities of the platform. Importantly, the team needs to have all the skills to deliver on the requirements, so it may start with three engineers and expand to keep up with demand. The platform owner will ensure the team size is adequate for agile delivery and create more teams to scale and broaden support.

IPE teams need to operate like an agile development team, so software engineering skills are critical to success. Skills like version control, code reviews, continuous integration/continuous delivery (CI/CD) and automated testing will be key to delivering reliable platforms. Platform engineers also will need to develop and maintain documentation, including platform solution documents, user manuals and runbooks (see Figure 5).

Figure 5: Infrastructure Platform Engineer Sample Skills

Infrastructure Platform Engineer Sample Skills



Source: Gartner
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Infrastructure platform engineers will apply software engineering practices to infrastructure automation to build self-service and API access to platform capabilities. The role requires continual learning to meet the evolving technology of the platform customers.

Develop Product Management and Software Engineering Skills

The new skills required for IPE can be divided into two buckets: product management and software engineering. In addition to the operations and engineering skills, I&O teams need to be successful and optimize the delivery of infrastructure.

Product management skills. Platforms need to be delivered as products, and users must be treated as customers. Platform teams need to be highly engaged with customers and prioritize new platforms and enhancements based on their needs. The platforms need to be demand-driven — they should begin simple and evolve based on feedback and adoption. Appointing a platform owner will be key to success as they will play the role of a product owner (see [Hire and Develop Product Managers Who Fit the 3 Stages of the Product Life Cycle](#)). The platform owner will ensure the team has the right mix of skills, including “people” skills (such as communication and collaboration) and business acumen.

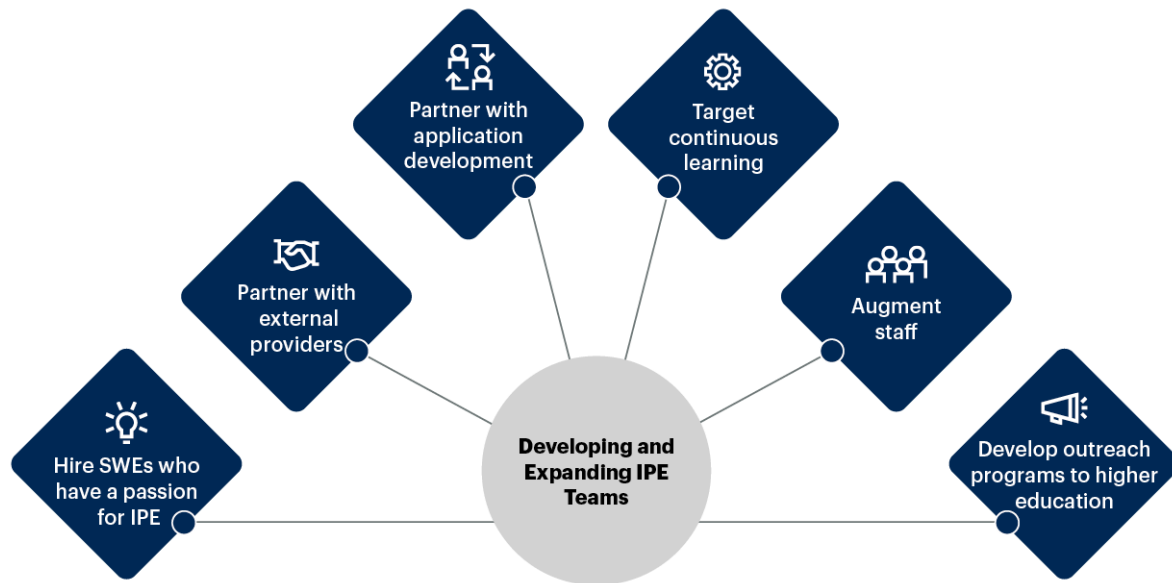
Software engineering skills. Modern infrastructure platforms require familiarity with technologies such as APIs, infrastructure as code (IaC), cloud-native infrastructure and composable architectures. The IPE team needs to acquire skills in software engineering to take existing coding and scripting skills and make them more efficient, repeatable and scalable. Software engineering practices need to be applied to all aspects of provisioning and operations of infrastructure platforms (see [Infographic: Build These Essential Skills to Secure Modern Software Development](#)).

The I&O leader must make skills development a priority by investing in learning and development plans for their IPE teams to fill skills gaps. Navigating digital learning content providers can be confusing and challenging. It may require multiple providers to build a targeted training for the technologies and practices to support the platform capabilities. I&O leaders can use the [Market Guide for Digital Learning Content Providers](#) to navigate the market.

Develop a Comprehensive Strategy That Includes Hiring, Training and Partnering With Service Providers

Adoption of modern, distributed architectural patterns of cloud-native infrastructure platforms means that developing and delivering software, data science and business solutions are more complex than ever before. The I&O leader needs to create a comprehensive strategy to continue to evolve their IPE teams to ensure they are equipped with the resources and capabilities they need to deliver value to their organization. This strategy needs to include all options described below (and encapsulated in Figure 6).

Figure 6: Strategies for Developing and Expanding Resources for an IPE Team

Strategies for Developing and Expanding Resources for an IPE Team

Source: Gartner
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Hire software engineers with a passion for IPE. Software engineering skills are critical to an IPE team's success, so recruiting software engineers who are interested in platform engineering should be a priority. Hire these software engineers at the inception of an IPE. These software engineers should have experience in agile product development.

Partner with external providers. Many service providers are offering platform engineering services. The I&O leader must consider bringing in a partner to help with accelerated adoption and ensure success with initial implementation of the IPE (see [Magic Quadrant for Custom Software Development Services, Worldwide](#)).

Partner with application development teams. Many organizations are investing in platform engineering to improve developer experience and reduce cognitive load due to the complexity of cloud-native infrastructure platforms. I&O leaders need to collaborate with stakeholders from software engineering and platform engineering, and enable contributions from the broader community to reduce duplication and improve value delivery.

Target continual learning. Investing in the continual training of staff is critical to success, as they need to adapt to the changing technology environment and evolving needs of customers. I&O leaders must invest in training — such as online courses, targeted training classes and conference attendance — to keep their teams learning and engaged. For example, Nationwide’s IT leaders focused on developing, not just hiring, the skills they need, communicating the organization’s changing needs to staff, and providing compelling incentives and avenues to develop those skills (see [Rebalanced Technical Skills Portfolio \(Nationwide\)](#)).

Augment staff. If hiring full-time equivalents (FTEs) is not an option, augment the team with external qualified resources. This will be beneficial to the IPE team’s success.

Develop outreach programs with higher education. The I&O leader should work with HR and select a few institutions known for engineering programs that align with the required IPE skills. Offer internships to developing software engineers to give them hands-on experience, and consider how these interns will fit into the organization’s growth and development plans.

Evidence

This research is based on analyst interactions with clients and industry knowledge.

Recommended by the Authors

Some documents may not be available as part of your current Gartner subscription.

[Case Study: Infrastructure Platform Experience Engineering \(Repsol\)](#)

[Case Study: Installing Agile Ways of Working in Infrastructure \(Audi\)](#)

[Top Reasons Why I&O Platform Engineering Efforts Fail](#)

[Designing a Platform-Oriented I&O Operating Model](#)

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